MOTORCYCLE FOOTPEG BRACKET

- **Problem:** "Motorcycle footpeg brackets are safety-critical components subjected to combined rider loads. This project applies topology optimisation to minimise mass while maintaining SF ≥ 2.5."
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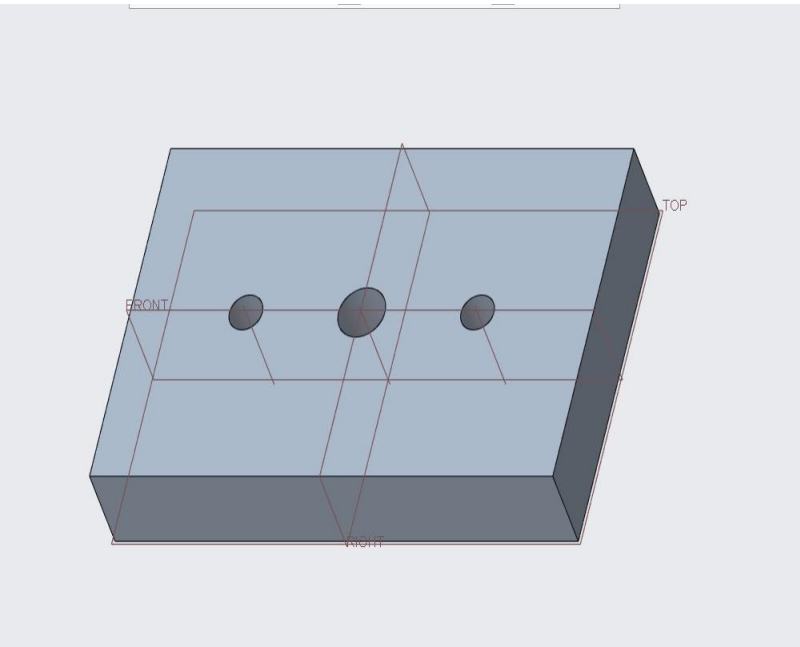
Requirements table:	Al 6061-T6	Yield: 276 MPa	Target SF: ≥ 2.5	Mass reduction: ≥ 40%

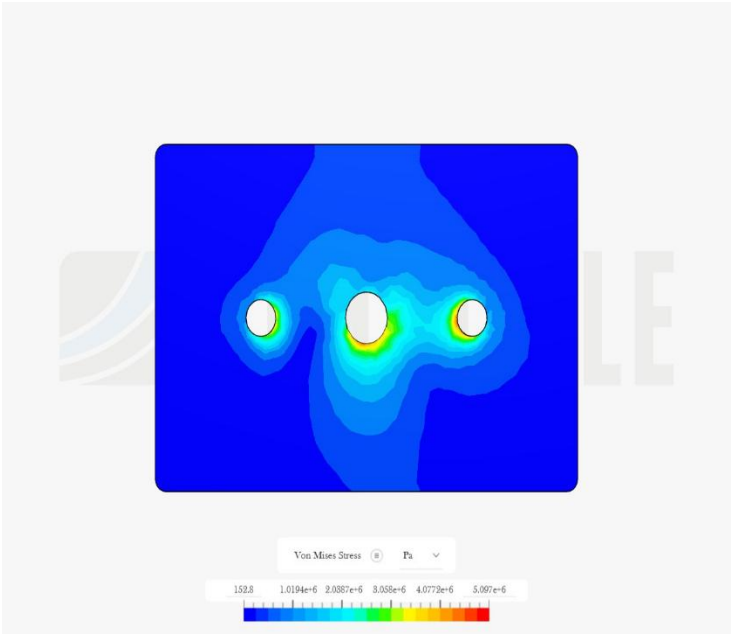
Load case:	80 kg rider	Vertical: 470 N	Braking: 314 N	Lateral: 235 N	Combined: 612 N

- **Process:**
  1. Bounding box CAD (Creo)
  2. FEA validation (Ansys + SimScale)
  3. Shape optimisation (Fusion 360)
  4. Engineering drawing (Creo)

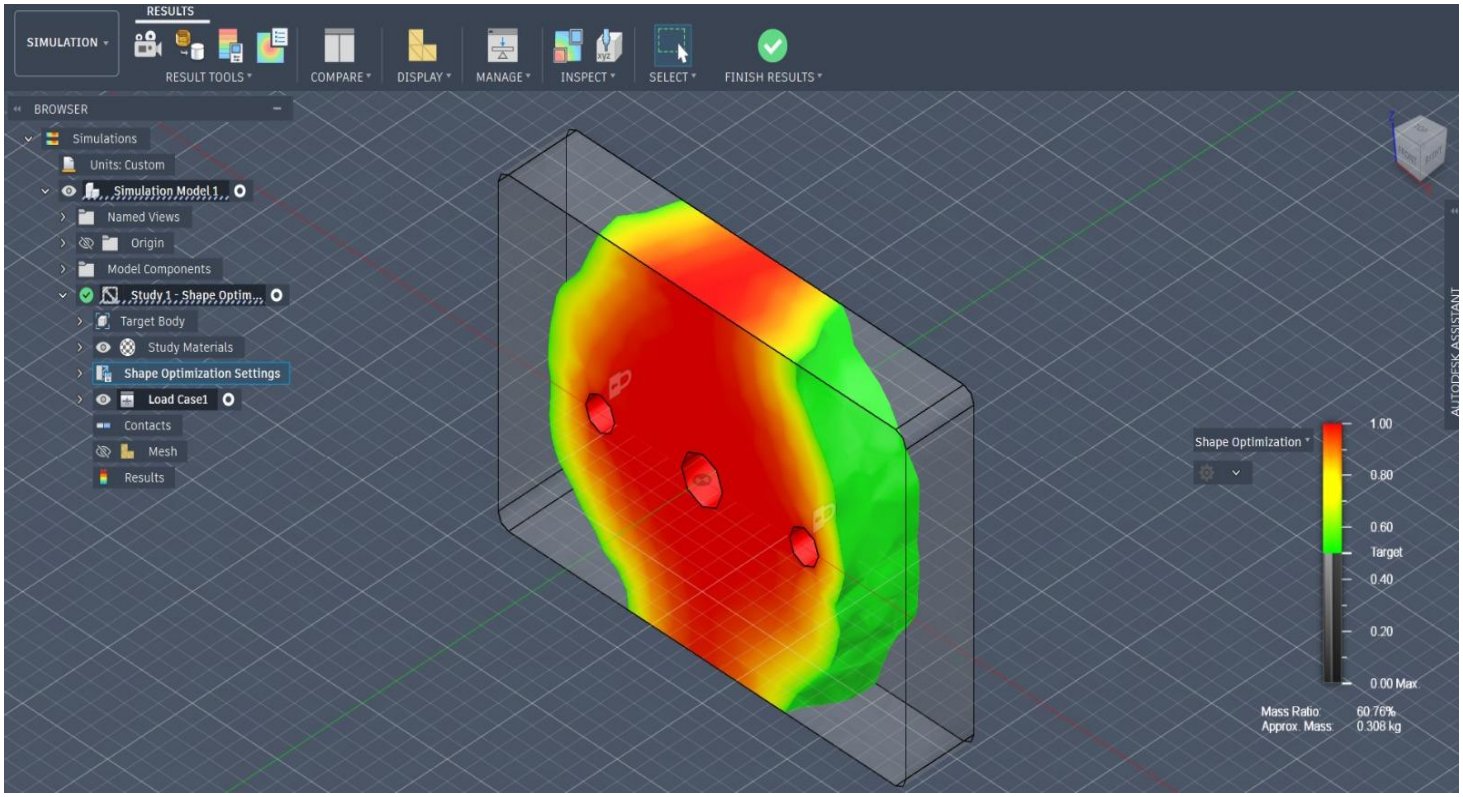
- **Results:**
  1. Original mass: 0.507 kg
  2. Optimised: 0.308 kg
  3. Reduction: 39.2%
  4. Max stress: 5.10 MPa
  5. SF: 54.1



Bounding box



Von Mises plot



Topology result